(a)

U1
DB15

RL
V6
V1
V5
LL
V4
LA
V3
RA
V2

AWDO
RLDOUT
IN1P
IN8P
IN7P
IN3P
IN6P
IN2P
IN5P
IN32N
IN4P

R6 22.1K
R24 10K
R1 22.1K
R16 10K
R3 22.1K
R2 10K
R5 22.1K
R4 10K
R8 22.1K
R7 10K
R17 22.1K
R9 10K
R19 22.1K
R18 10K
R21 22.1K
R20 10K
R23 22.1K
R22 10K
R10 22.1K
R14 10K

C37 47pF
C1 47pF
C2 47pF
C28 47pF
C4 47pF
C3 47pF
C6 47pF
C5 47pF
C22 47pF
C21 47pF
C29 47pF
C23 47pF
C31 47pF
C30 47pF
C33 47pF
C32 47pF
C35 47pF
C34 47pF
C36 47pF
C24 47pF

D1 D7
D2 D8
D3 D9
D4 D10
D5 D11
D6 D12
D14 D13
D16 D15
D18 D17
D20 D19

1N4148WS

AGND

(b)

Arduino R3 Header
U2

RESET
3V
5V
GND1
VIN
A1
A2
A3
A4SDA
A5SCL

SCL
SDA
AREF
GND
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1
D0

5V_ST
+Vbat
PWDN
RESET
DOUT
DIN

CLKSEL
SCLK
DAISY_IN
CS
DRDY

R12 10K
R13 10K
R15 10K

AGND
DVDD

(c)

S2
SWITCH-DPDTAYZ0202
5V_ST
5V_Reg
5V_ST
+Vbat
AWDO
Vin_3.3V_Reg

R11 330
LED1
AWDO
AGND

JST XH 3
JP1
Vbat_in
AGND

DGND
AGND

+Vbat
Vbat_in
L1 3.3uH
U4
LP2985AIM5-5.0
VIN
ON/OFF#
GND
VOUT
BYPASS
L3 3.3uH
5V_Reg

C42 10uF
C45 1uF
C44 10nF
C39 10uF
C38 10uF

AGND

Vin_3.3V_Reg
U3
LP2985A-33DBVR
VIN
GND
ON/OFF#
VOUT
BYPASS
L2 3.3uH
DVDD

C18 1uF
C16 10nF
C19 10uF
C43 10uF

DGND

(d)

ADS1198

IN8N
IN8P
IN7N
IN7P
IN6N
IN6P
IN5N
IN5P
IN4N
IN4P
IN3N
IN3P
IN2N
IN2P
IN1N
IN1P
TESTP_PACE_OUT1
TESTN_PACE_OUT2
AVDD
AVSS
AVDD
AVDD
AVSS
VREFP
VREFN
VCAP4
NC
VCAP1
NC
VCAP2
RESV1
AVSS

IN
IN
IN
IN
IN
IN
IN32N
IN3P
IN2P
IN
IN1P
TESTP
TESTN
AWDO
AWDO
AWDO
VERFP
VCAP4
VCAP1
VCAP2

WCT
RLDOUT
RLDIN
RLDINV
RLDREF
AVDD
AVSS
AVSS
AVDD
VCAP3
AVDD1
AVSS1
CLKSEL
DGND
DVDD
DGND
DVDD
DRDY
GPIO4
GPIO3
GPIO2
DOUT
GPIO1
DAISY_IN
SCLK
CS
START
CLK
RESET
PWDN
DIN
DGND

AWDO
AWDO
VCAP3
AWDO
CLKSEL
DVDD
DVDD
DRDY
GPIO4
GPIO3
GPIO2
DOUT
GPIO1
DAISY_IN
SCLK
CS
CLK
RESET
PWDN
DIN

AGND
AGND
DGND
DGND

(e)

VCAP4 VCAP3 VCAP2 VCAP1 VERFP

C17 1uF
C7 1uF
C8 100nF
C9 1uF
C10 22uF
C13 10uF
C12 100nF

AGND

AWDO
AWDO
DVDD

C25 1uF
C26 100nF
C20
C15 100nF
C41 1uF
C40 100nF

AGND
AGND
DGND

WCT
IN
C11 100pF
H1
2pinsbarrette
TESTP
TESTN

AGND

(f)

J1
RLDOUT
DVDD
RLDIN
C14 1nF
R25 670K
C27 NI
E/D
GND
VDD
OUT
U5 NI
CLK
RLDINV
DGND